ALA2: Python Deep Dive: Interview Intensive

Topic name: Accessing List Elements

- **Name of topic points:**
  - Using Index (Positive Indexing),
  - Using Negative Indexing,
  - Accessing a Range (Slicing)
  - Slicing with Step,
  - Accessing Entire List, and
  - Using Loop to Access Elements.

Q1. How do you access the first element of a list?

- **Theory:** Use index 0 to access first element
- **Syntax:** list[0]
- **Example:** [10,20,30][0] → 10

2. How do you access the last element?

- **Theory:** Use negative index -1
- **Syntax:** list[-1]
- **Example:** [10,20,30][-1] → 30

3. What is negative indexing?

- **Theory:** Access elements from the end of list
- **Syntax:** list[-index]
- **Example:** [1,2,3][-2] → 2

4. What will be the output?

"a = [5, 10, 15]
print(a[1])"

- **Theory:** Index starts from 0
- **Answer:** 10

5. What is slicing in lists?

- **Theory:** Access a range of elements
- **Syntax:** list[start:end]
- **Example:** [1,2,3,4][1:3] → [2,3]

6. What will be the output?

“a = [1,2,3,4]
print(a[:2])”

- **Theory:** Default start = 0
- **Answer:** [1,2]

7. What will be the output?

“a = [1,2,3,4]
print(a[2:])”

- **Theory:** Default end = last
- **Answer:** [3,4]

8. What is slicing with step?

- **Theory:** Skip elements using step
- **Syntax:** list[start:end:step]
- **Example:** [1,2,3,4,5][::2] → [1,3,5]

9. What will be the output?

“a = [10,20,30,40]
print(a[::2])”

- **Theory:** Step = 2 skips one element
- **Answer:** [10,30]

10. What does list[:] do?

- **Theory:** Returns full copy of list
- **Syntax:** list[:]
- **Example:** [1,2][:] → [1,2]

11. What will be the output?

“a = [1,2,3]
print(a[-2])”

- **Theory:** Negative index from end
- **Answer:** 2

12. How to access elements of nested list?

- **Theory:** Use multiple indexes
- **Syntax:** list[i][j]
- **Example:** [[1,2],[3,4]][1][0] → 3

13. What will be the output?

```
a = [[1,2],[3,4]]
print(a[0][1])
```

- **Theory:** First select list, then element
- **Answer:** 2

14. What happens if index is out of range?

- **Theory:** Raises IndexError
- **Example:** [1,2][5] → Error

15. How to safely access list elements?

- **Theory:** Check length before accessing
- **Example:**

```
if i < len(a):
    print(a[i])
```